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Physics-Constrained Deep Learning for Permeability Field Data Assimilation: A U-Net-Based Surrogate Enhanced with Attention Mechanisms

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Abstract

Accurate reconstruction of subsurface permeability fields is critical for reliable reservoir forecasting and decision-making. However, conventional numerical simulators are computationally expensive, and standard data-driven approaches often fail to capture the strong heterogeneity present in real reservoirs. This study proposes a physics-constrained deep learning framework for efficient permeability field data assimilation, integrating an attention-enhanced U-Net surrogate model with partial differential equation (PDE) constraints and evolutionary optimization.

Permeability fields are generated via sequential Gaussian simulation (SGS), while a Differential Evolution (DE) algorithm is employed to assimilate them using sparse observations from five wells, including permeability and bottom-hole flow rate measurements. The U-Net architecture facilitates hierarchical feature extraction, and attention modules dynamically emphasize spatial dependencies between permeability and pressure distributions. In addition, the integration of PDE constraints ensures that the learned pressure field adheres to governing flow physics. Additionally, an adaptive learning rate strategy is introduced to improve convergence efficiency and model robustness.

Benchmark comparisons indicate that the proposed PDE-constraint surrogate model improves pressure prediction accuracy by 87.88% relative to baseline architectures without attention and physics constraints. The end-to-end framework reduces computational cost to 8% of conventional simulators, while maintaining relative pressure prediction errors below 3% across diverse geological scenarios. Results highlight the model's capacity to capture fine-scale heterogeneities, enhance physical fidelity, and accelerate the data assimilation process, offering a scalable solution for high-resolution reservoir characterization.

Keywords: Physics-constrained deep learning, U-Net surrogate model, Permeability data assimilation, Attention mechanism, Differential Evolution, Reservoir heterogeneity

Introduction

Accurate characterization of subsurface permeability fields is fundamental to reservoir simulation, production forecasting, and decision-making in petroleum engineering. As a critical reservoir property, permeability governs the flow behavior of fluids within porous media, and its spatial heterogeneity significantly impacts reservoir performance and associated uncertainties. (Evensen et al. 2007) However, generating high-resolution permeability maps remains a formidable challenge due to limited well control, complex geological heterogeneity, and the high computational cost of conventional inversion techniques. (Oliver et al. 2018; Remy et al. 2009)

Traditional inversion methods typically integrate numerical reservoir simulators with optimization algorithms to assimilate production and pressure data. While these methods are grounded in physical principles and provide robust predictions, they are computationally intensive, particularly when using commercial simulators. (Liu et al. 2005; Silva et al. 2017) Thousands of forward simulations are often required to perform history matching or uncertainty quantification, resulting in prohibitive computational burdens for large-scale or high-resolution models. (wang et al. 2025)

Recently, physics-constrained data-driven surrogate models have emerged as a promising alternative due to their superior computational efficiency. Once trained, these models can perform predictions at a fraction of the time required by traditional simulators, often as low as 8% of the original computational cost. (Bi et al. 2024; Liu et al. 2024) Unlike purely data-driven approaches, which depend heavily on large volumes of labeled data that are often unavailable in real-world applications, physics-constrained models incorporate governing physical laws, typically represented by elliptic or parabolic PDEs derived from Darcy's law and mass conservation, into the learning process via loss function regularization or soft constraints. This allows them to achieve satisfactory predictive performance even with limited labeled samples. (Nagao et al. 2024)

Meanwhile, advances in deep learning have further enabled the extraction of complex, nonlinear relationships in high-dimensional systems. Convolutional neural networks (CNNs), particularly encoder-decoder architectures such as U-Net, have shown strong potential in predicting spatially distributed subsurface properties. (Han et al. 2025; Wang et al. 2025) However, most existing models treat the permeability-to-pressure mapping as a purely data-driven task, disregarding the underlying flow physics. (wang N et al, 2021b)(Xu R et al, 2022) As a result, such models often yield predictions that violate physical consistency, produce chaotic or noisy outputs, and lack interpretability from a hydrogeological standpoint. Furthermore, the strong spatial heterogeneity typical of subsurface formations poses additional challenges to standard CNNs, which struggle to extract meaningful multi-scale features in such complex environments. (Xu R, 2021)

To overcome these limitations, this study proposes a novel physics-constrained deep learning framework comprising three key innovations:

- **Attention-Enhanced U-Net Surrogate:** A U-Net architecture integrated with spatial attention mechanisms to dynamically capture multi-scale correlations between pressure and permeability distributions.
- **PDE-Based Physics Regularization:** Incorporation of flow-governing PDE constraints into the learning process to enforce physical consistency in surrogate model predictions.
- **Evolutionary Data Assimilation:** Integration of a Differential Evolution (DE) algorithm to perform efficient permeability inversion using sparse and noisy well observations.

The framework is trained and validated on permeability fields generated via Sequential Gaussian Simulation (SGS), with synthetic observations derived from five wells, including both permeability and

bottom-hole pressure measurements. To simulate realistic conditions, Gaussian noise with a standard deviation of 3% is added to the observed data. The inversion results are quantitatively evaluated by comparing the reconstructed permeability fields to the true fields. In Scenario 1, the average relative error is 4.61%, while Scenario 2 yields 6.96%, indicating that the proposed framework provides accurate and robust inversion performance under noisy, sparse observation conditions.

The remainder of this paper is organized as follows: Section 2 presents the physics-informed U-Net architecture with attention enhancements. Section 3 introduces the data assimilation strategy and the implementation of the DE algorithm. Section 4 describes the experimental design and evaluation metrics. Section 5 discusses the results and insights. Finally, Section 6 concludes the study and outlines future research directions.

Method

physics-informed U-Net architectures with attention enhancements

To improve the accuracy and physical consistency of permeability field inversion under sparse observations, we develop a deep learning framework based on a physics-informed U-Net architecture augmented with attention mechanisms. This section outlines the governing physical equations, the integration of physics into the neural network, and the design of the model architecture.

Governing Equations of Single-Phase Flow. The fundamental physical constraint governing pressure propagation in porous media is the single-phase Darcy flow equation under slightly compressible fluid assumptions. (Baird et al. 2005) The flow is governed by the following elliptic partial differential equation:

$$\nabla \cdot \left(-\frac{k(\mathbf{x})}{\mu} \nabla p(\mathbf{x}) \right) = q(\mathbf{x}), \mathbf{x} \in \Omega \quad (1)$$

where: k_x is the permeability field, mD; μ is the fluid viscosity, mPa·s; p_x is the pressure field, MPa; q_x is the source/sink term, m³/day; Ω represents the spatial domain.

For well modeling, we incorporate the *Peaceman* well model to relate grid-based pressure to well rates. (Peaceman D W, Chen Z) The flow rate at well block iii is computed by:

$$Q_i = \frac{2\pi k_i h_i}{\mu \ln \left(\frac{r_{e,i}}{r_{w,i}} \right) + s_i} (p_i - p_{wf,i}) \quad (2)$$

Where, h_i is the formation thickness, m; r_e is the effective drainage radius, m; r_w is the wellbore radius, m; s_i is the skin factor; p_i is the block pressure, Mpa; p_{wf} is the bottom-hole flowing pressure, Mpa.

The finite-difference form of the governing PDE for single-phase fluid flow is expressed as:

$$\frac{\partial}{\partial x} \left(\frac{K_x}{B_o \mu_o} \frac{\partial p}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{K_y}{B_o \mu_o} \frac{\partial p}{\partial y} \right) + q_o = \frac{\phi C_o}{B_o} \frac{\partial p}{\partial t} \quad (3)$$

where B_o is the oil formation volume factor, rm^3/sm^3 ; μ_o is the oil viscosity, $mPa \cdot s$; C_o denotes the oil compressibility coefficient, $1/Mpa$; p represents the reservoir pressure, MPa; K_x and K_y denote the permeabilities in the x and y directions, mD; ϕ denotes the reservoir porosity, and q_o is the oil production rate at standard conditions, m³/day.

Physics-Informed Loss Function. To embed the flow physics into the learning process, the U-Net model is trained using a composite loss function that combines data fidelity and physics consistency. (Rai et al. 2020; Willard et al. 2020) The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{PDE_constraint}} \mathcal{L}_{\text{PDE_constraint}} \quad (4)$$

$\mathcal{L}_{\text{data}}$ is the observation loss, typically measured as the mean squared error (MSE) between predicted and observed pressures at well locations.

$L_{\text{PDE_constraint}}$ is the PDE residual loss, ensuring the predicted pressure field satisfies the governing flow equation in a weak form (e.g., using finite differences or automatic differentiation).

λ_{data} and $\lambda_{\text{PDE_constraint}}$ are weighting coefficients that balance the two components.

This hybrid loss function guides the neural network to produce pressure fields that not only match observed values but also adhere to physical laws of fluid flow.

U-Net with Attention Mechanisms

The U-Net architecture serves as the backbone of the proposed surrogate model due to its ability to perform dense predictions over spatial domains with limited labeled data. Originally designed for biomedical image segmentation, U-Net features an encoder–decoder structure with skip connections that retain high-resolution spatial information while extracting hierarchical features through convolutional layers. (Chen et al. 2007; Suri et al. 2023) This design is particularly advantageous in permeability-to-pressure mapping tasks, where spatial resolution and multi-scale feature interactions are critical. The algorithmic structure of the U-Net with attention mechanisms is illustrated in Fig. 1.

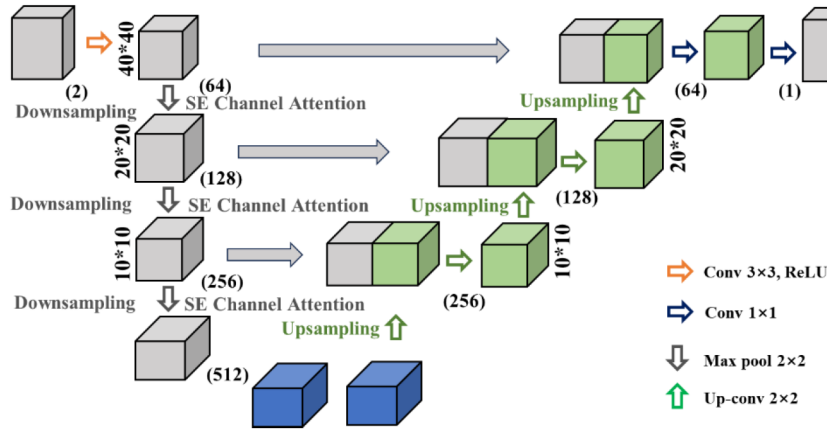


Figure 1—The structure of U-Net with Attention Mechanisms

Encoder–Decoder Structure. The encoder progressively downsamples the input permeability field through a sequence of convolutional blocks, each consisting of convolution, batch normalization, and ReLU activation layers, followed by max-pooling for spatial compression. This process extracts abstract features at multiple resolutions. (Liu et al. 2024) Conversely, the decoder upsamples the encoded features using transposed convolutions (or bilinear interpolation) and fuses them with corresponding high-resolution features from the encoder via skip connections. This ensures that both global structural patterns and local details are preserved in the output pressure field.

Let E_i and D_i represent the feature maps at the i -th encoder and decoder level, respectively. The skip connection passes E_i to D_i such that:

$$D_i = \text{DecoderBlock}(D_{i+1}) + \text{Attention}(E_i) \quad (5)$$

where, the attention module selectively filters and enhances encoder features before fusion.

Squeeze-and-Excitation (SE) Channel Attention. Although skip connections enable the model to recover spatial information lost during the encoding process, they may also transmit irrelevant or noisy features, which can be particularly detrimental in heterogeneous reservoirs. (Hu et al. 2020) To address this issue, we incorporate two types of attention mechanisms—spatial and temporal (as illustrated in Fig. 2)—into the skip connections to enhance adaptivity in both spatial and channel dimensions.

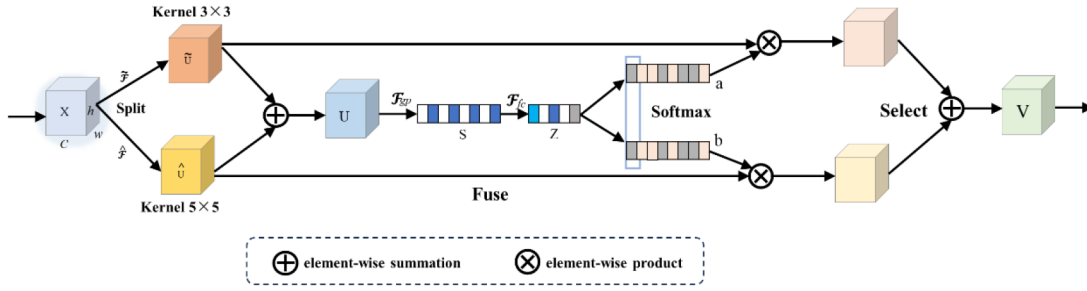


Figure 2—Spatial and temporal attention mechanisms

(a) Channel attention

Channel attention captures the interdependencies between feature channels, allowing the network to prioritize more informative features. For each encoder output $E \in \mathbb{R}^{C \times H \times W}$, a global average pooling is first applied to generate a channel descriptor $z \in \mathbb{R}^C$:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W E_{c,i,j} \quad (6)$$

where, $z_{(c)}$ represents the result of global average pooling for the c -th channel; $E_{c,i,j}$ is the encoder output feature map, which is a 3-dimensional tensor with dimensions $C \times H \times W$.

Then, a two-layer fully connected network (FC-ReLU-FC-Sigmoid) is applied to produce attention weights $\alpha \in [0,1]^C$, which are used to scale the input feature map:

$$\tilde{E}_c = \alpha_c \cdot E_c \quad (7)$$

where, $\tilde{E}_c$ represents the scaled feature map for the c -th channel; $\alpha_{(c)}$ is the attention weight for the c -th channel, It is a value in the range $[0, 1]$ (as $\alpha \in [0,1]^C$); E_c denotes the original feature map of the c -th channel before scaling.

This emphasizes channels relevant to fluid flow patterns, such as high-permeability streaks or pressure anomalies.

(b) Spatial Attention

While channel attention focuses on "what" to attend to, spatial attention addresses "where" to attend. It generates a spatial attention map by applying average and max pooling along the channel dimension, concatenating the results, and applying a convolutional filter:

$$M_s = \sigma(f^{7 \times 7}([AvgPool(E); MaxPool(E)])) \quad (8)$$

$$\tilde{E} = M_s \odot E \quad (9)$$

where, M_s represents the spatial attention map. σ denotes the Sigmoid activation function. $f^{7 \times 7}$ stands for a convolutional operation with a 7×7 filter. $AvgPool(E)$ Performs global average pooling on the input feature map E along the channel dimension. $MaxPool(E)$ Carries out global max pooling on the input feature map E along the channel dimension; E is the original feature map output by the encoder.

This helps the model focus on spatially important regions—such as near-well zones or permeability discontinuities—that have a stronger influence on pressure propagation.

The attention-augmented U-Net is trained under the guidance of physics-informed loss functions described in Section 2.2. The attention modules assist the network in learning spatial patterns that are physically consistent, especially in areas where high gradient pressures or source/sink terms occur. For instance, in regions surrounding production wells, the attention weights dynamically increase to focus the model's learning capacity on accurately resolving local pressure drops governed by Darcy's law and the Peaceman well model.

In addition, the attention-enhanced architecture improves model robustness under data-sparse conditions, a common challenge in practical reservoir scenarios where permeability measurements are limited to a few well locations. By adaptively emphasizing the most informative spatial and channel features, the network avoids overfitting to noise or irrelevant geological structures, thus achieving better generalization to unseen permeability realizations.

Network Architecture. The complete U-Net with attention mechanisms is composed of: **Input layer:** 2D permeability fields ($k(x, y)$); **Encoder path:** 4 convolutional blocks with downsampling; **Attention skip connections:** channel + spatial attention modules between encoder and decoder levels; **Decoder path:** 4 convolutional blocks with upsampling; **Output layer:** predicted pressure field ($p(x, y)$). Detailed network parameters are provided in Table 1.

Table 1—Network Architecture of the U-Net Attention-Based Surrogate Model

Part	Layers	Output size
Encoder	Input data	[32, 2, 40, 40]
	Downsample 1	[32, 64, 40, 40]
	MaxPool2D	[32, 64, 20, 20]
	Downsample 2	[32, 128, 20, 20]
	ChannelAttention(128)	[32, 128, 20, 20]
	MaxPool2D	[32, 128, 10, 10]
	Downsample 3	[32, 256, 10, 10]
	ChannelAttention(256)	[32, 256, 10, 10]
	Downsample 4	[32, 256, 5, 5]
	ChannelAttention(512)	[32, 512, 5, 5]
Decoder	ConvTranspose2D	[32, 256, 10, 10]
	Concatenate with encoder features	[32, 512, 10, 10]
	Upsample	[32, 256, 10, 10]
	Upsample	[32, 128, 20, 20]
	Concatenate with encoder features	[32, 256, 20, 20]
	Upsample	[32, 128, 20, 20]
	ConvTranspose2D	[32, 64, 40, 40]
	Concatenate with encoder features	[32, 128, 40, 40]
	Upsample	[32, 64, 40, 40]
	Output data	[32, 1, 40, 40]
ChannelAttention	AdaptiveAvgPool2D	
	AdaptiveMaxPool2D	
	FC1, 16 neurons	
	ReLU	
	FC2, 16 neurons	
	Sigmoid	

All components are implemented using PyTorch and trained using the Adam optimizer with a hybrid loss combining data observation error and physics-based PDE residuals.

In this study, we propose a physics-informed U-Net architecture enhanced with attention mechanisms to improve pressure field predictions under sparse permeability observations. The framework integrates fundamental single-phase flow physics, including Darcy's law and the Peaceman well model, into a hybrid loss function that guides the network toward physically consistent predictions. The U-Net backbone, with its encoder–decoder structure and skip connections, is augmented by channel and spatial attention modules to selectively emphasize critical features in both the spatial and channel dimensions. These attention mechanisms enhance the model's capability to capture heterogeneity-induced flow patterns and improve robustness in data-limited scenarios. The entire model is implemented in PyTorch, trained using a physics-informed composite loss, and its detailed architecture—including layer-wise configurations—is tabulated for reproducibility.

Data Assimilation Strategy and Differential Evolution Implementation

Building upon the physics-informed surrogate model presented in Section 2, this section introduces the inverse modeling framework and the associated optimization strategy for reconstructing permeability fields from sparse observations. Accurate and efficient assimilation of limited reservoir data is critical for recovering high-resolution permeability distributions, especially in highly heterogeneous formations. To address this challenge, we adopt a data assimilation approach that integrates the physics-constrained surrogate with a global optimization algorithm—Differential Evolution (DE)—to infer permeability structures based on sparse pressure measurements from well locations. (Jeyakumar et al. 2014; Qing et al. 2006)

Problem Setup and Observation Configuration. The computational domain represents a two-dimensional single-layer porous reservoir. The underlying permeability field is characterized as a log-normally distributed random field, generated via Sequential Gaussian Simulation (SGS). The goal of the inversion is to reconstruct the spatial distribution of permeability, $k(x, y) \in \mathbb{R}$, based on sparse and noisy measurements obtained from a limited number of monitoring wells.

At each monitoring location, two types of measurements are assumed to be available: (1) local permeability values and (2) bottom-hole flowing pressures. To reflect realistic measurement uncertainty, zero-mean Gaussian noise with a standard deviation of 3% is added to both the permeability and pressure data.

Inverse Modeling Formulation. The inversion process is cast as a high-dimensional optimization problem. The goal is to identify the optimal permeability field k^* such that the pressure field predicted by the surrogate model matches the observed bottom-hole pressures, and the local permeability values at well locations are consistent with the known measurements. (Han et al. 2025) Furthermore, the reconstructed pressure field is required to satisfy the governing physical laws described in Section 2.1.

The cost function $J(k)$ to be minimized is defined as:

$$J(\mathbf{k}) = \lambda_{\text{data}} \sum_{i=1}^{N_{\text{wells}}} \left(\|k(x_i, y_i) - k_{\text{obs}}(x_i, y_i)\|^2 + \|p_{\text{sim}}(x_i, y_i) - p_{\text{wf},i}\|^2 \right) + \lambda_{\text{phys}} \mathcal{L}_{\text{PDE}} \quad (10)$$

where, $p_{\text{sim}}(x, y)$ is the pressure field predicted by the surrogate model, bar; $\mathcal{L}_{\text{PDE}}$ is the residual loss enforcing consistency with the single-phase flow equation; λ_{data} and λ_{phys} are weighting factors controlling the trade-off between data fidelity and physical consistency.

This hybrid formulation allows for robust inversion of permeability distributions in the presence of limited and noisy data while ensuring physical plausibility of the results.

Differential Evolution for Permeability Inversion. Given the high-dimensional and non-convex nature of the inverse problem, we employ Differential Evolution (DE) as a global optimization strategy. DE is a population-based, derivative-free evolutionary algorithm that is particularly well-suited for geophysical parameter estimation problems. The specific algorithmic structure is shown in Figure 3.

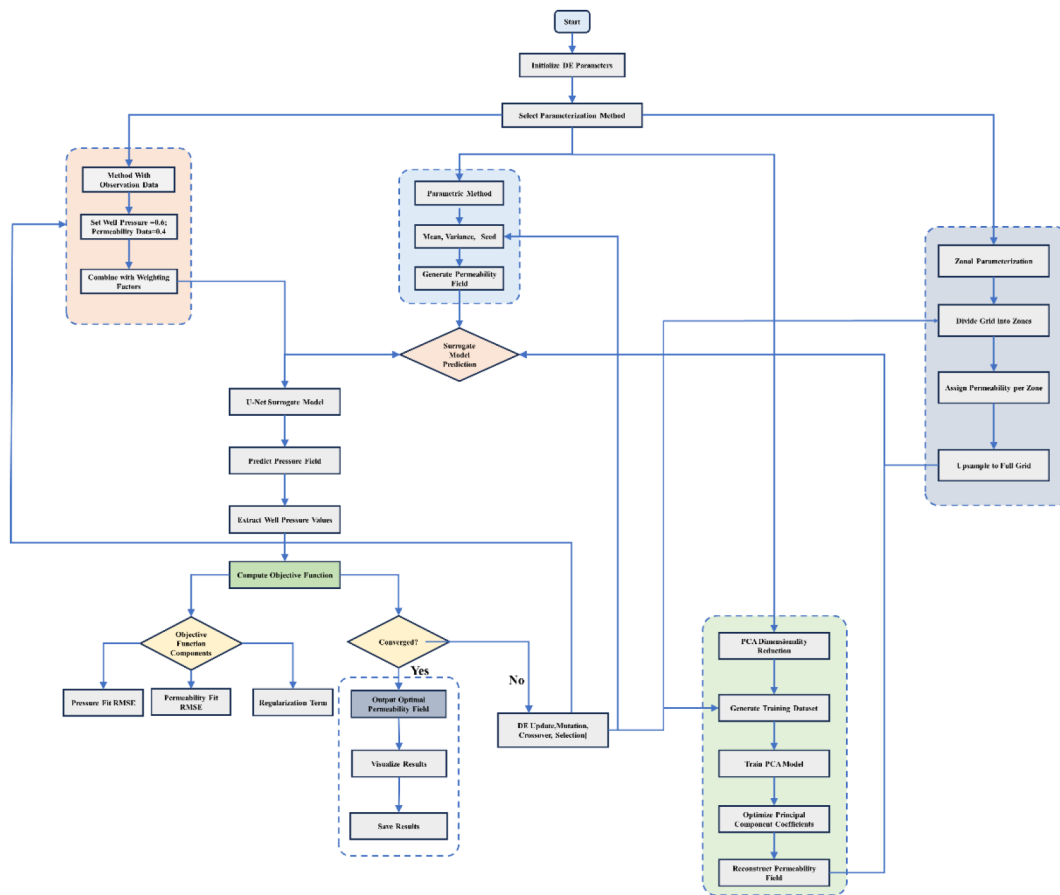


Figure 3—Differential Evolution (DE) optimization strategy

The algorithm workflow consists of Initialization, Mutation, Crossover, and Selection.

①**Initialization:** A population of candidate permeability fields $\{k_j\}_{j=1}^{N_p}$ is randomly sampled within predefined log-permeability bounds. Each candidate represents a flattened 40×40 grid.

②**Mutation:** For each target vector k_i , a donor vector is generated using:

$$v_i = k_{r1} + F \cdot (k_{r2} - k_{r3}) \quad (11)$$

where k_{r1}, k_{r2}, k_{r3} are distinct randomly selected individuals from the population, and F is the mutation factor.

③**Crossover:** A trial vector u_i is formed by combining elements of the donor and target vectors through binomial crossover:

$$u_{i,d} = \begin{cases} v_{i,d} & \text{if } \text{rand}(0, 1) < CR \text{ or } d = d_{\text{rand}} \\ k_{i,d} & \text{otherwise} \end{cases} \quad (12)$$

Where, $u_{i,d}$ is the trial vector formed for the i -th target vector; $v_{i,d}$ Denotes the d -th element of the donor vector v_i ; CR is the crossover rate and d_{rand} is a randomly selected dimension to ensure diversity.

④**Selection:** The objective function J is evaluated for both u_i and k_i . The individual with the lower cost proceeds to the next generation:

$$k_i^{(t+1)} = \begin{cases} u_i & \text{if } J(u_i) < J(k_i) \\ k_i & \text{otherwise} \end{cases} \quad (13)$$

Where, $k_i^{(t+1)}$ is the candidate solution for the i -th individual in the next generation (denoted as $t + 1$, where t is the current generation); u_i denotes the trial vector. k_i Represents the original target vector (or the original candidate solution) for the i -th individual in the current generation of the population.

⑤**Termination:** The optimization terminates once the convergence criteria are satisfied or a maximum number of generations (typically 300~500) is reached.

This section presents a physics-constrained inverse modeling framework that leverages a surrogate-assisted data assimilation strategy to infer heterogeneous permeability fields from sparse and noisy well measurements. The underlying permeability distribution is modeled as a log-normally distributed random field over a 2D porous domain. Observations consist of local permeability values and bottom-hole pressures at a limited number of monitoring wells, with Gaussian noise (3%) added to emulate realistic uncertainty. The inversion task is formulated as a high-dimensional optimization problem, balancing data fidelity and physical consistency via a hybrid loss function that integrates observational mismatch and PDE residuals. To solve this non-convex problem efficiently, Differential Evolution (DE) is employed as a derivative-free global optimizer. DE iteratively updates a population of candidate permeability fields through mutation, crossover, and selection steps, aiming to minimize the hybrid cost function. The optimization continues until convergence or a predefined maximum number of generations is reached. This integrated framework enables robust reconstruction of fine-scale reservoir heterogeneity even under data-sparse conditions.

Experimental Setup and Evaluation Metrics

This section outlines the experimental configuration used to validate the proposed permeability inversion framework. We describe the generation of synthetic datasets, the training of the physics-informed surrogate model, the implementation details of the differential evolution optimizer, and the metrics employed to assess inversion accuracy and computational efficiency.

Synthetic Dataset and True Model Generation

To benchmark the inversion performance under controlled conditions, a synthetic reference model was constructed using **Sequential Gaussian Simulation (SGS)** with a predefined variogram model.(Müller et al. 2022) The true log-permeability field is defined over a 40×40 grid, yielding 1600 unknowns. The permeability values follow a log-normal distribution with a mean of $\ln(k)=4.0$ and standard deviation of 1.0 (in log-scale), and spatial continuity is imposed via an exponential variogram with a correlation length of 10 grid blocks in both directions.

Given this reference field, the pressure solution is computed by numerically solving the single-phase flow equation under specified boundary and source conditions (as described in Section 2.1). The resulting pressure distribution and permeability values at the five well locations (see Section 3.1) are extracted to serve as observations. To emulate realistic field measurements, **Gaussian noise ($\sigma = 3\%$)** is added independently to both pressure and permeability values at observation points.(Sedarat and Fisher 2008)

Training of the Surrogate Model

The surrogate model used in this study is a U-Net architecture enhanced with attention modules and trained in a physics-informed manner. The training dataset consists of **1,100 synthetic permeability–pressure pairs** generated via high-fidelity simulations under varied permeability realizations. The model is trained to minimize a hybrid loss function combining: **Data loss:** Pointwise error between predicted and simulated pressures; **Physics loss:** Residuals of the governing PDE computed via automatic differentiation.

The model is trained using the Adam optimizer with an initial learning rate of 1×10^{-4} , batch size of 64, and early stopping based on validation loss. Training is conducted on a single NVIDIA 4090Ti GPU and converges in approximately 1 hour (3504.9 seconds). Once trained, the model can infer pressure fields for arbitrary permeability inputs in under 50 milliseconds per sample.

Differential Evolution Implementation

The Differential Evolution (DE) algorithm was implemented with the hyperparameters summarized in Table 2. A log-uniform sampling strategy was employed to initialize the permeability population across several orders of magnitude. At each iteration, the trained surrogate model was used to evaluate candidate solutions, with convergence determined by either a stagnation criterion (relative objective change < 0.01 for 30 generations) or a maximum generation limit of 300.

Table 2—Hyperparameters and Settings for Differential Evolution (DE) Optimization

Parameter	Value	Description
Population size	30	Number of candidate solutions per generation
Mutation factor, FFF	0.8	Controls amplification of differential variation
Crossover rate, CRCRCR	0.9	Probability of crossover between donor and target
Maximum number of generations	300	Optimization halts after this number is reached
Permeability search space	[10 ⁻¹ , 10 ³]	Log-uniform range for initial population sampling
Initial sampling strategy	Uniform in log-space	Ensures wide coverage across several magnitudes
Evaluation model	Trained surrogate model	Used to predict pressure and compute objective value
Convergence criterion	Relative change < 0.01 for 30 generations	Early stopping condition

Evaluation Metrics

To quantitatively assess the performance of the inversion strategy, (Zhao et al. 2025; Kong et al. 2024) we report the following metrics:

Root Mean Square Error (RMSE) of the reconstructed permeability field:

$$RMSE_k = \sqrt{\frac{1}{N} \sum_{i=1}^N ((k_i^{\text{inv}}) - (k_i^{\text{true}}))^2} \quad (14)$$

Average Relative Error (ARE) in the reconstructed permeability field:

$$ARE_k = \frac{1}{N} \sum_{i=1}^N \left| \frac{k_i^{\text{inv}} - k_i^{\text{true}}}{k_i^{\text{true}}} \right| \quad (15)$$

Forward model consistency, measured as the average PDE residual norm over the entire domain, ensures that the predicted pressure fields satisfy physical constraints.

Computation time, measured both for total inversion runtime and average surrogate model inference time.

Additionally, visual comparison of reconstructed permeability fields against the ground truth is presented to qualitatively assess spatial resolution and feature recovery.

Results and Discussion

Building upon the methods described in Sections 2–4, This section presents the results of a comprehensive evaluation of the proposed physics-constrained surrogate framework for permeability field data assimilation. The evaluation focuses on surrogate model accuracy, assimilation effectiveness, computational efficiency, and the interpretability enabled by attention mechanisms and PDE constraints.

Surrogate Model Performance and Comparison

The reference pressure fields used for training and validation were generated from a single-phase reservoir flow simulator based on Darcy’s law and the Peaceman well model. The simulation was conducted

on a 40×40 Cartesian grid, with each grid block measuring $20m \times 20m \times 20m$. The basic reservoir and fluid properties are summarized in Table 3. Five vertical wells were strategically positioned to capture representative geological variability, with their spatial coordinates defined as follows: Well 1 – (10,10), Well 2 – (20,20), Well 3 – (30,30), Well 4 – (10,30), and Well 5 – (30,10). Constant bottom-hole pressures were imposed at production wells, while injection wells operated under fixed-rate conditions to ensure stable flow patterns during the simulation.

Table 3—Reservoir and Fluid Properties Used in the Reference Numerical Simulation

Parameter	Value	Unit	Description
Grid size	40×40	—	Cartesian grid
Grid block dimensions	$20 \times 20 \times 20$	m	$\Delta x, \Delta y, \Delta z$
Initial reservoir pressure	300	bar	Uniform initial pressure
Porosity, ϕ	0.20	—	Matrix porosity
Permeability range	50 – 1000	mD	Heterogeneous field generated stochastically
Oil viscosity, μ_o	6.0	mPa·s	Single-phase oil
Oil compressibility, C_o	1.0×10^{-4}	1/bar	Slightly compressible fluid
Formation compressibility, C_u	1.0×10^{-4}	1/bar	Slightly compressible rock
Formation volume factor, B_o	1.1	-	At reference conditions
Well radius, r_w	0.10	m	Used in Peaceman well model
Well bottom-hole pressure	250	bar	Fixed BHP for producers
Simulation time	40	days	Quasi-steady flow field generation

Representative examples of the surrogate model predictions are illustrated in Figure 4 and Figure 6 for the PDE-constrained U-Net with attention mechanisms, and in Figure 5 and Figure 7 for the baseline U-Net with attention mechanisms. The results (Sample 60, 80) demonstrate that the proposed physics-constrained surrogate accurately reproduces fine-scale pressure variations, particularly in regions with sharp permeability contrasts and strong pressure gradients. In contrast, the baseline U-Net tends to oversmooth localized features, resulting in a loss of critical flow details essential for characterizing heterogeneous formations.

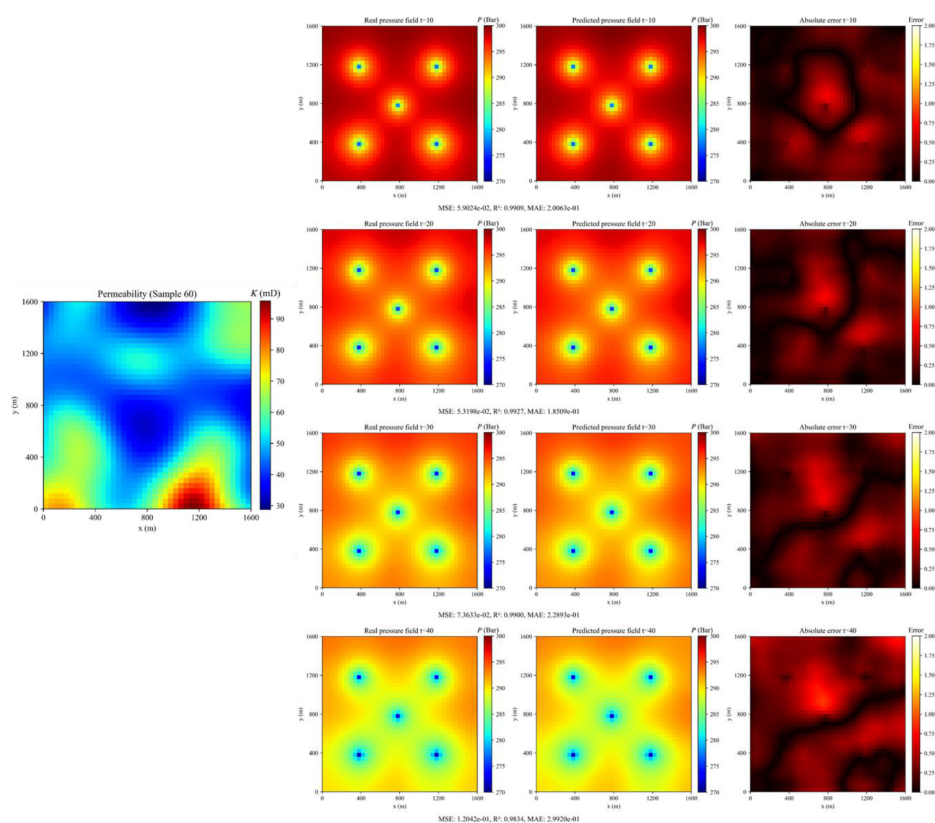


Figure 4—Predicted pressure field using a PDE-constrained U-Net model with embedded attention mechanisms. (Sample 60)

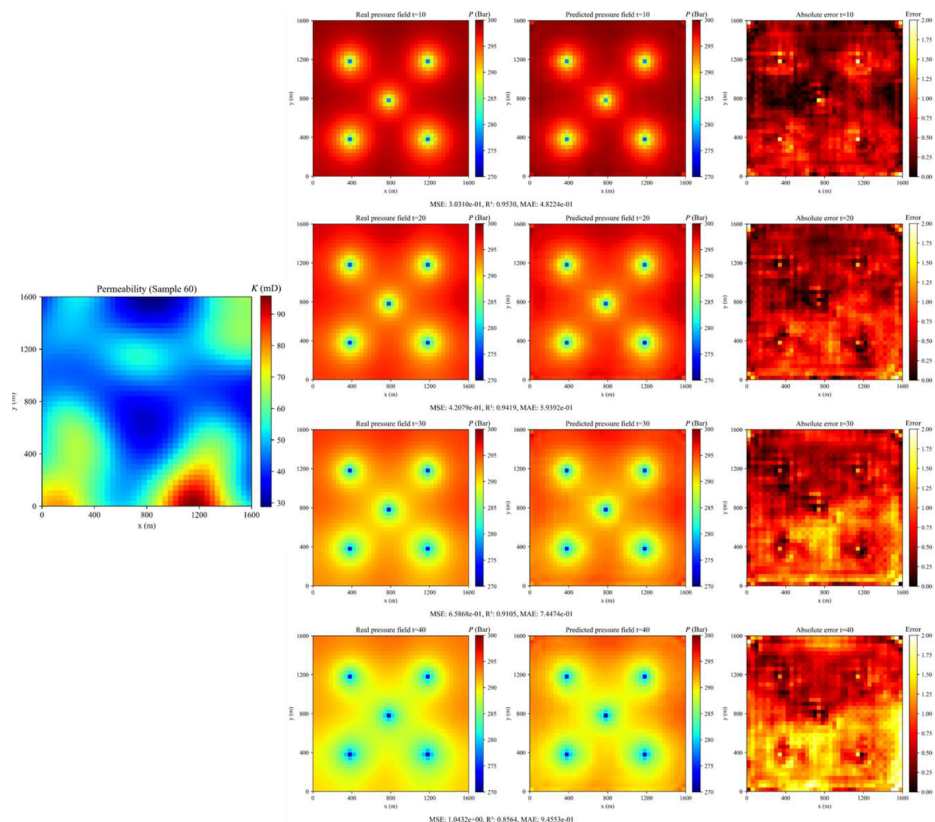


Figure 5—Predicted pressure field using a U-Net model with embedded attention mechanisms. (Sample 60)

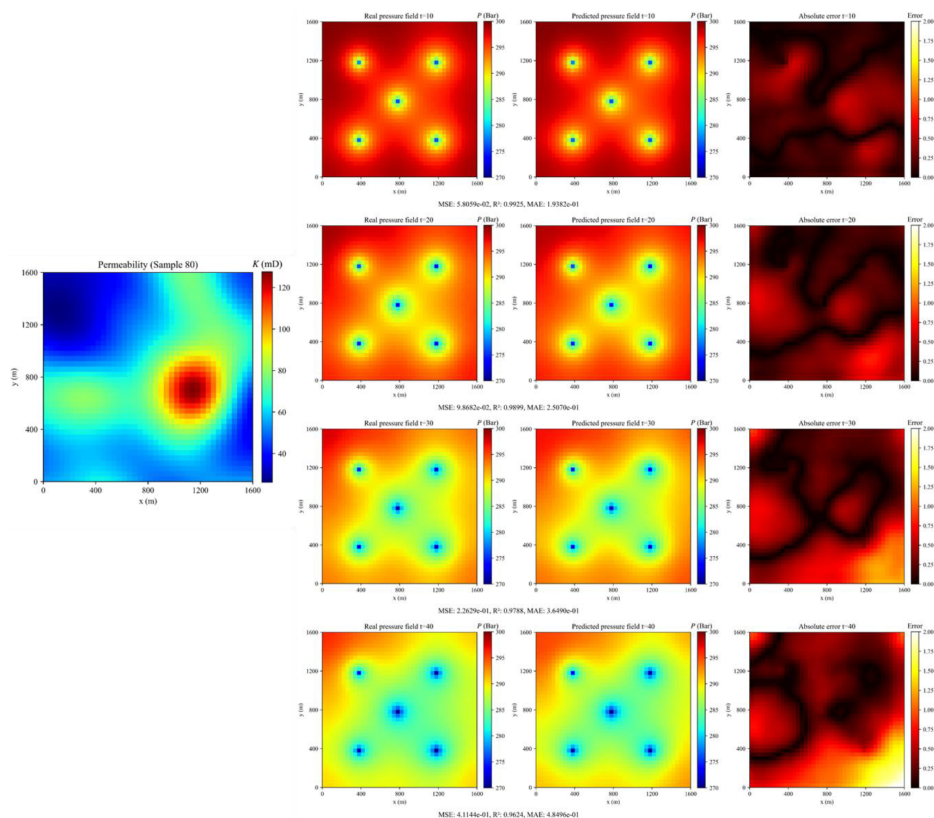


Figure 6—Predicted pressure field using a PDE-constrained U-Net model with embedded attention mechanisms. (Sample 80)

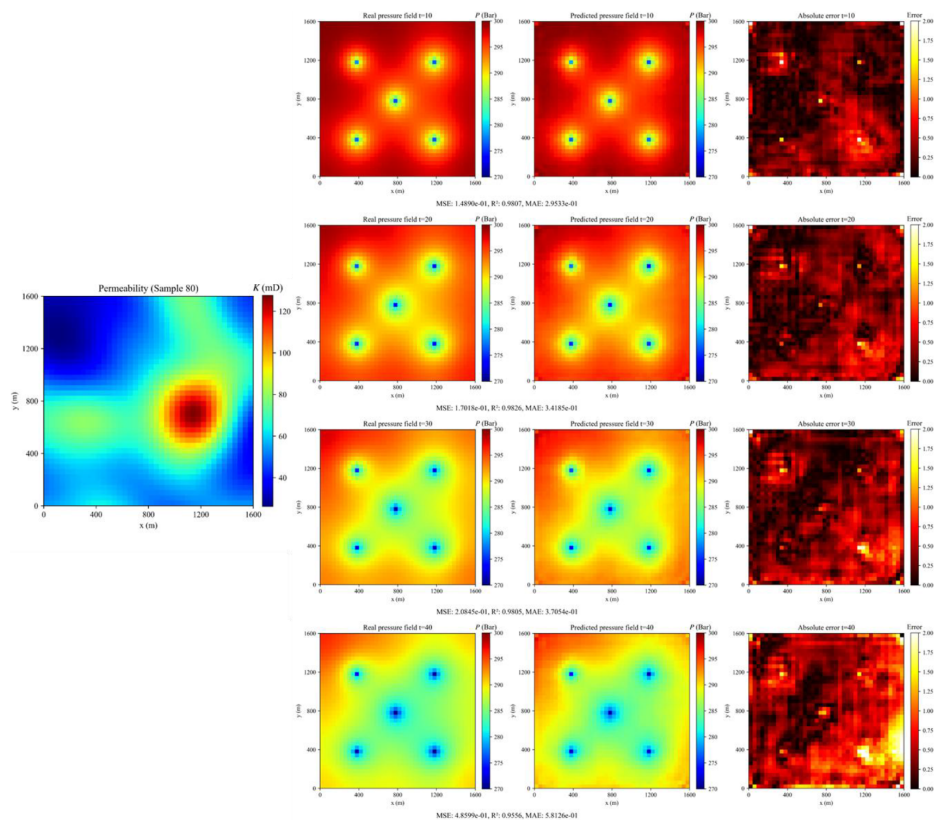


Figure 7—Predicted pressure field using a U-Net model with embedded attention mechanisms. (Sample 80)

Table 4 presents a quantitative performance comparison between the proposed model and the standard U-Net. The proposed model achieves an 87.88% reduction in mean square error (MSE), indicating significantly improved pressure prediction accuracy. The improvement is particularly evident in bottomhole pressure predictions, where the incorporation of PDE-based physical constraints leads to notably higher accuracy compared to purely data-driven models. This enhanced performance directly translates into more reliable bottomhole flow rate calculations, which are highly sensitive to pressure deviations. The performance gain is especially pronounced in regions with strong heterogeneity, where conventional data-driven architectures often underperform due to spatial averaging effects.

Table 4—The Performance evaluation of a PDE-constrained U-Net model with embedded attention mechanisms

Permeability field sample number		Evaluation indicator			
	T (step)	Model	MSE	MAE	R ²
60 Sample	t = 10 steps	40-step data-driven model	0.30	0.48	0.95
		a PDE-constrained U-Net model with embedded attention mechanisms	0.06	0.20	0.99
	t = 20 steps	40-step data-driven model	0.42	0.59	0.94
		a PDE-constrained U-Net model with embedded attention mechanisms	0.05	0.19	0.99
	t = 30 steps	40-step data-driven model	0.66	0.74	0.91
		a PDE-constrained U-Net model with embedded attention mechanisms	0.08	0.23	0.99
80 Sample	t = 40 steps	40-step data-driven model	1.04	0.95	0.86
		a PDE-constrained U-Net model with embedded attention mechanisms	0.12	0.30	0.98
	t = 10 steps	40-step data-driven model	1.48	0.30	0.98
		a PDE-constrained U-Net model with embedded attention mechanisms	0.06	0.19	0.99
	t = 20 steps	40-step data-driven model	0.17	0.34	0.98
		a PDE-constrained U-Net model with embedded attention mechanisms	0.10	0.25	0.99
	t = 30 steps	40-step data-driven model	0.21	0.37	0.98
		a PDE-constrained U-Net model with embedded attention mechanisms	0.23	0.36	0.98
	t = 40 steps	40-step data-driven model	0.48	0.58	0.95
		a PDE-constrained U-Net model with embedded attention mechanisms	0.41	0.48	0.96

Building on these improvements, the integration of attention mechanisms further enhances the model’s capability by directing computational focus toward geologically significant regions. This targeted representation contributes to improved spatial resolution and fidelity in pressure prediction. These findings underscore the effectiveness of combining physics-informed constraints with attention modules in surrogate modeling for subsurface flow problems. Notably, the model maintains high predictive performance without relying on large-scale labeled datasets. By embedding PDE constraints, the model can infer physically consistent pressure distributions even in unobserved regions, thereby mitigating overfitting and improving generalizability across diverse geological realizations.

Inversion Results via DE-Based Data Assimilation

The DE-based data assimilation procedure, using sparse measurements from five wells, successfully reconstructed permeability fields that are consistent with both the prior geostatistical models (SGS) and the observed dynamic flow responses. To better reflect field conditions, 3% Gaussian noise was added to the observed pressures and permeability values at the five wells. Figures 8 and 9 present the bottomhole pressure results after 40 assimilation steps using different reference permeability fields. The final pressure distributions closely match the observed values, demonstrating the effectiveness and correctness of the assimilation process. This indicates that the inverse solution satisfies both physical laws and observational constraints. Figures 10 and 11 illustrate the reconstructed permeability fields obtained through the DE-based assimilation, including the reference (true) permeability, the predicted permeability, and the absolute error map. The visual agreement between the reconstructed and true fields further validates the model's ability to infer high-fidelity spatial distributions from limited and noisy observations.

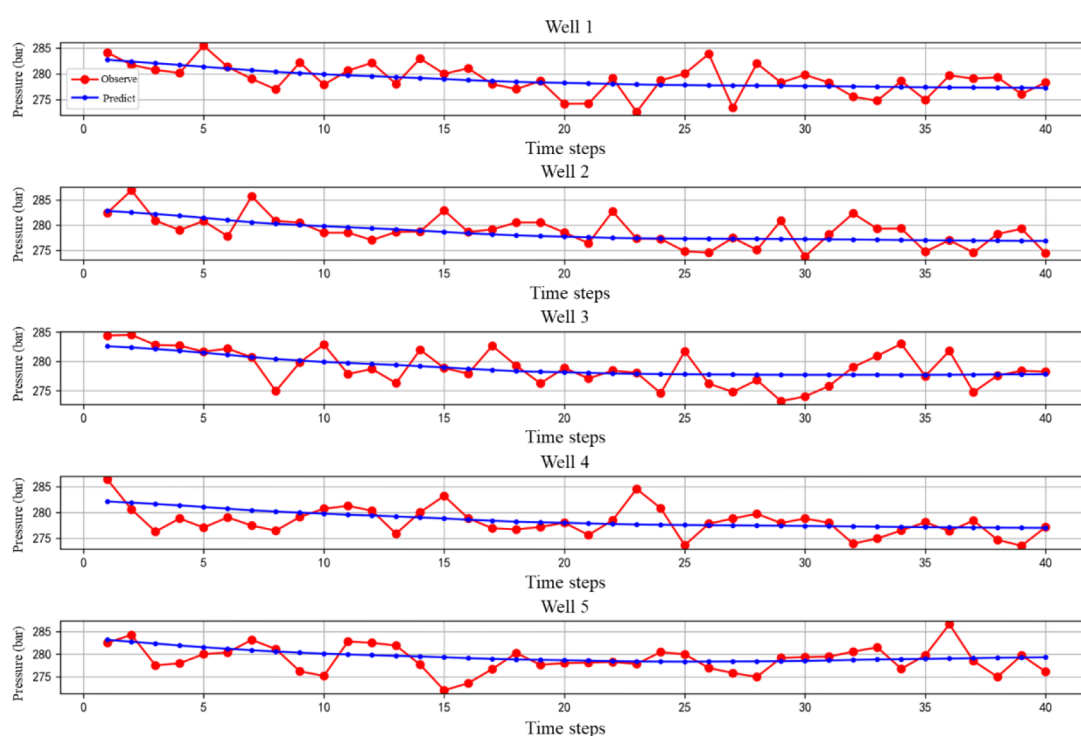


Figure 8—Bottomhole pressure results after 40 assimilation steps (Example 1)

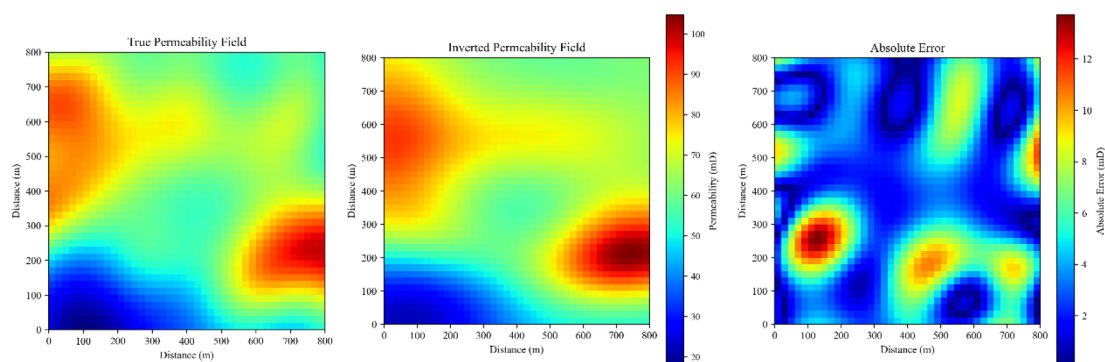


Figure 9—Inverted permeability field and absolute error map (Example 1)

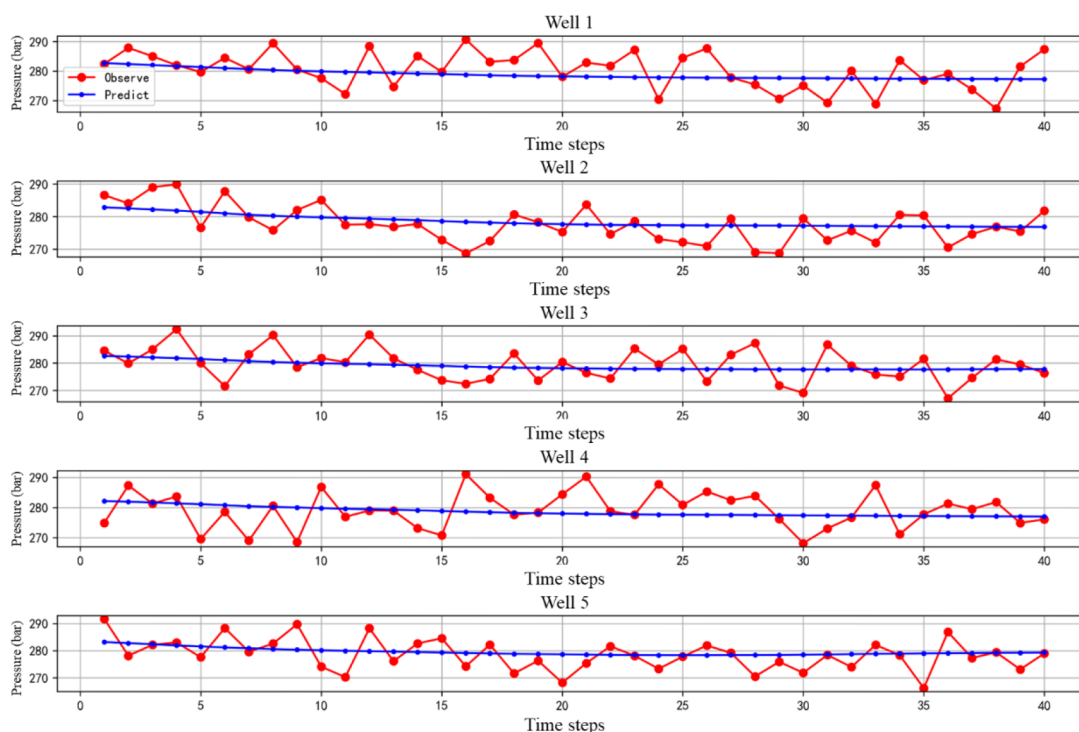


Figure 10—Bottomhole pressure results after 40 assimilation steps (Example 2)

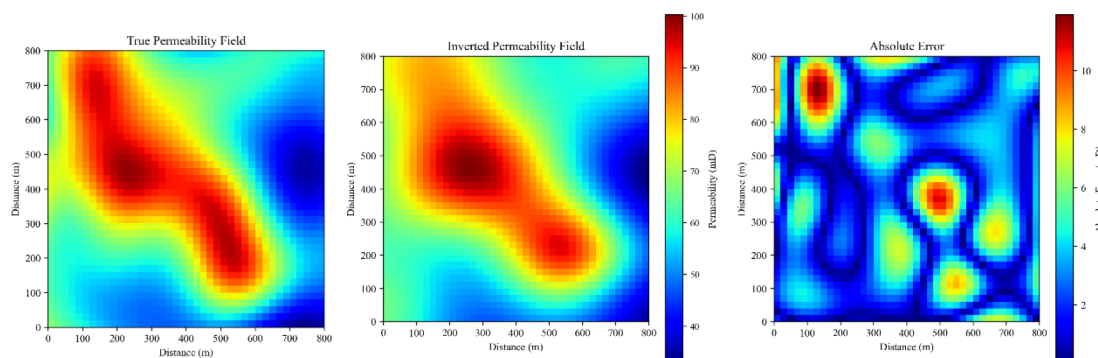


Figure 11—Inverted permeability field and absolute error map (Example 2)

The results indicate that the assimilation of permeability field data for Example 1 resulted in an average relative error of 6.96%, while for Example 2, the average relative error was 4.61%. Moreover, the spatial correlation structures in the predicted permeability fields remained geologically plausible, preserving the continuity and anisotropy imposed by the prior SGS realizations. This suggests that the framework not only fits the data but also maintains geological realism.

In practical reservoir workflows, computational speed is critical. The proposed surrogate model achieved **an order-of-magnitude reduction in computation time**, consuming only 8% of the time required by conventional numerical solvers. This efficiency enables high-throughput analysis and opens the door for real-time applications such as production optimization, scenario screening, or ensemble-based uncertainty quantification.

Discussion on Results

The ablation experiments demonstrate the complementary roles of the embedded attention mechanism and the PDE-based physical constraints. The attention mechanism improves the model's ability to localize fine-scale features, which is particularly valuable in regions with sharp pressure gradients or abrupt permeability

contrasts. In contrast, the PDE constraints enforce global physical consistency by acting as an inherent regularization term, effectively preventing nonphysical pressure predictions and ensuring adherence to underlying conservation laws.

When the attention module was removed, the model's performance in heterogeneous regions noticeably declined. Specifically, it tended to underestimate local pressure variations near impermeable barriers or high-permeability channels due to insufficient focus on critical spatial features. Conversely, when the PDE constraints were excluded, the model occasionally generated pressure fields that violated fundamental physical principles, such as monotonicity or mass conservation. The combined integration of attention and PDE constraints thus provided a robust balance: the attention mechanism offered local adaptability, while the physics constraints guaranteed global stability—both essential qualities for reliable data assimilation in geologically complex reservoirs.

Beyond the overall performance improvements, several nuanced insights emerged during the experiments:

①Boundary sensitivity: Slightly elevated prediction errors were observed near domain boundaries, especially in cases with sparse measurement coverage. This indicates that future work could benefit from incorporating boundary-aware embeddings, improved padding strategies, or learning-based boundary condition treatments to better constrain edge behavior.

②Uncertainty quantification: Although the deterministic differential evolution (DE) algorithm proved effective for the current scope, extending the framework with Bayesian inference or ensemble-based methods could provide richer estimates of posterior uncertainty, particularly in ill-posed or data-scarce regions where multiple permeability realizations may fit the same sparse observations.

③Model scalability: Tests conducted on higher-resolution grids confirmed that the training time scales approximately linearly with the problem size, which supports the approach's feasibility for larger-scale reservoir simulations. However, practical deployment at field scale may require strategies to manage GPU memory and computational resource demands, such as model parallelism or multi-GPU implementations.

④Real-field applicability: While the current experiments are based on synthetic data, the framework's embedded physics-awareness and demonstrated generalizability indicate promising potential for real-field applications. With high-quality field measurements, minimal architectural modifications would be needed to adapt the model for practical reservoir characterization, closed-loop production optimization, or time-lapse monitoring.

Overall, this study demonstrates the practical viability of integrating physics constraints and attention mechanisms into deep learning surrogates for reservoir data assimilation. The proposed framework strikes an effective balance between physical fidelity, predictive accuracy, and computational efficiency, making it particularly valuable for time-sensitive reservoir engineering tasks such as production forecasting, closed-loop field development, and probabilistic scenario analysis. By enabling rapid yet reliable assimilation of sparse field data, the framework facilitates informed decision-making under geological uncertainty and offers a scalable foundation for future extensions to more complex physics, including multiphase flow, thermal effects, and coupled geomechanics—thereby enhancing its applicability and impact in real-world reservoir management workflows.

Conclusion

This study presents a physics-constrained surrogate modeling framework that combines a PDE-constrained U-Net architecture with embedded attention mechanisms for efficient and reliable reservoir data assimilation. Based on the results and discussion, the following conclusions can be drawn:

1. The proposed framework, incorporating both spatial attention and PDE-informed loss, significantly improves pressure prediction accuracy over standard data-driven U-Net baselines. The attention

modules enhance feature localization in heterogeneous formations, while the physics constraints enforce global flow consistency.

2. The surrogate model accurately reconstructed permeability fields from sparse well observations, with average relative errors around 5.8%. It achieves high-resolution pressure prediction in under 50 milliseconds per sample.
3. Compared to conventional simulators, the framework reduces computational cost by approximately 92%, requiring only ~8% of the original runtime. This enables efficient deployment in workflows with limited data or real-time constraints.
4. The demonstrated scalability and generalization capabilities across different geological realizations indicate strong potential for real-field applications. With minimal modifications, the model can support tasks such as closed-loop production optimization, uncertainty quantification, scenario ranking, and time-lapse reservoir monitoring.
5. While this study primarily focuses on synthetic benchmarks, the methodology is inherently adaptable to field datasets, as the evolution of permeability fields aligns with geological structural changes. Future work will emphasize validation on field-scale cases to assess robustness under practical data limitations and operational uncertainties, thereby bridging the gap toward real-world deployment.

Reference

- Baird, J. and Dawson, C. 2005. The representer method for data assimilation in single-phase Darcy flow in porous media. *Computational Geosciences* **9** (4): 247–271.
- Bi, J. F., Li, J., Wu, K. L. et al 2024. A Physics-Informed Spatial-Temporal Neural Network for Reservoir Simulation and Uncertainty Quantification. *SPE JOURNAL* **29** (4): 2026–2043.
- Chen, J. N., Mei, J. R., Li, X. H. et al 2024. TransUNet: Rethinking the U-Net architecture design for medical image segmentation through the lens of transformers. *Medical Image Analysis* **97**. 103280.
- Chen Z. Reservoir simulation: mathematical techniques in oil recovery[M]. *Society for Industrial and Applied Mathematics*, 2007.
- Evensen, G., Hove, J., Meisingset, H. C. et al 2007. Using the EnKF for Assisted History Matching of a North Sea Reservoir Model. Paper presented at the SPE Reservoir Simulation Symposium. <https://doi.org/10.2118/106184-MS>.
- Hu, J., Shen, L., Albanie, S. et al 2020. Squeeze-and-Excitation Networks. *IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE* **42** (8): 2011–2023.
- Han, Y. F., Hamon, F. P., and Durlowsky, L. J. 2025. Accelerated training of deep learning surrogate models for surface displacement and flow, with application to MCMC-based history matching of CO₂ storage operations. *Geoenergy Science and Engineering* **246**. 213589.
- Jeyakumar, G. and Velayutham, C. S. 2014. Distributed heterogeneous mixing of differential and dynamic differential evolution variants for unconstrained global optimization. *SOFT COMPUTING* **18** (10): 1949–1965.
- Kong, Y. L., Zhang, J., Zhang, X. S. et al 2024. Applicability and limitation of compost maturity evaluation indicators: A review. *Chemical Engineering Journal* **489**. 151386.
- Liu, N. and Oliver, D. S. 2005. Ensemble Kalman filter for automatic history matching of geologic facies. *Journal of Petroleum Science and Engineering* **47** (3–4): 147–161.
- Liu, A. D., Li, J., Bi, J. F. et al 2024. A novel reservoir simulation model based on physics informed neural networks. *PHYSICS OF FLUIDS* **36** (11). 116617.
- Liu, D. B., Zhang, T. Q., Christensen, M. G. et al 2024. Multi-layer encoder-decoder time-domain single channel speech separation. *PATTERN RECOGNITION LETTERS* **181**: 86–91.
- Müller S, Schüller L, Zech A, et al GSTools v1. 3: a toolbox for geostatistical modeling in Python[J]. *Geoscientific Model Development*, 2022, **15**(7): 3161–3182.
- Nagao, Masahiro, Datta-Gupta, Akhil, Onishi, Tsubasa, and Sankaran, Sathish. 2024. Physics Informed Machine Learning for Reservoir Connectivity Identification and Robust Production Forecasting. *SPE Journal* **29** (09): 4527–4541. <https://doi.org/10.2118/219773-PA>.
- Oliver, Dean S. and Alfonzo, Miguel. 2018. Calibration of imperfect models to biased observations. *Computational Geosciences* **22** (1): 145–161. <https://doi.org/10.1007/s10596-017-9678-4>.
- Peaceman DW. Interpretation of well-block pressures in numerical reservoir simulation (includes associated paper 6988) [J]. *Society of Petroleum Engineers Journal*, 1978, **18**(03): 183–194.

- Qing, A. Y. 2006. Dynamic differential evolution strategy and applications in electromagnetic inverse scattering problems. *IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING* **44** (1): 116–125.
- Rai, R. and Sahu, C. K. 2020. Driven by Data or Derived Through Physics? A Review of Hybrid Physics Guided Machine Learning Techniques With Cyber-Physical System (CPS) Focus. *IEEE Access* **8**: 71050–71073. <https://doi.org/10.1109/ACCESS.2020.2987324>.
- Remy, N., Boucher, A., Wu, J., 2009. Applied geostatistics with SGeMS: A user's guide. Cambridge University Press.
- Suri, J. S., Bhagawati, M., Agarwal, S. et al 2023. UNet Deep Learning Architecture for Segmentation of Vascular and Non-Vascular Images: A Microscopic Look at UNet Components Buffered With Pruning, Explainable Artificial Intelligence, and Bias. *IEEE ACCESS* **11**: 595–645.
- Sedarat, H. and Fisher, K. 2008. Multicarrier communication in presence of biased-Gaussian noise sources. *Signal Processing* **88** (7): 1627–1635.
- Silva, V. L. S., Emerick, A. A., Couto, P., and Alves, J. L. D. 2017. History matching and production optimization under uncertainties Application of closed-loop reservoir management. *Journal of Petroleum Science and Engineering* **157**: 860–874.
- Wang, N., Chang, H., Zhang, D., 2021b. Theory-guided auto-encoder for surrogate construction and inverse modeling. *Comput. Methods Appl. Mech. Eng.* **385**, 114037 <https://doi.org/10.1016/j.cma.2021.114037>.
- Wang, N. Z. and Durlafsky, L. J. 2025. Deep learning framework for history matching CO2 storage with 4D seismic and monitoring well data. *GEOENERGY SCIENCE AND ENGINEERING* **248**. 213736.
- Willard, J., Jia, X., Xu, S. et al 2020. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Computing Surveys* **55** (4): 1–37. <https://doi.org/10.1145/3514228>.
- Xu, R., Zhang, D. X., and Wang, N. Z. 2022. Uncertainty quantification and inverse modeling for subsurface flow in 3D heterogeneous formations using a theory-guided convolutional encoder-decoder network. *Journal of Hydrology* **613**. 128321.
- Xu, R., Zhang, D., Rong, M., Wang, N., 2021b. Weak form theory-guided neural network (TgNN-wf) for deep learning of subsurface single- and two-phase flow. *J. Comput. Phys.* **436**, 110318 <https://doi.org/10.1016/j.jcp.2021.110318>.
- Zhao, Beichen, Yao, Yuedong, Xiao, Zheyu et al 2025. A Fusion Model for Predicting the Minimum Miscible Pressure of a CO₂–Oil System Based on a Classification Sampling Strategy. *Energy & Fuels*. <https://doi.org/10.1021/acs.energyfuels.5c01119>.